

HDOC Day-36

Menu-Driven String Operations in C

Problem Statement

Write a menu-driven C program using separate functions to: (1) count characters in a string without using a built-in length function, (2) print each character of a string on a new line, (3) count vowels and consonants in a string, and (4) exit.

Algorithm

1. Display the menu repeatedly until Exit is selected.
2. Choice 1: read a complete line, traverse it until the null character, count the characters, and display the count.
3. Choice 2: read a complete line and traverse it; print each character on a separate line.
4. Choice 3: read a complete line; for each alphabetic character, convert it to lowercase for checking. Count a, e, i, o, u as vowels and the remaining alphabetic characters as consonants. Ignore spaces, digits, and punctuation.
5. Choice 4: terminate the program. For any other choice, display an invalid-choice message.
6. Use separate functions for the three operations. Do not use strlen() for Choice 1.

Test Case Table

TC	Choice	Input	Expected Result
1	1	String: Hello	Number of characters = 5
2	1	String: Hello World	Number of characters = 11 (space included)
3	2	String: CAT	Prints C, A and T on separate lines
4	2	String: A B	Prints A, space and B on separate lines
5	3	String: Education	Vowels = 5, Consonants = 4
6	3	String: C Programming 123!	Vowels = 3, Consonants = 9; non-letters ignored
7	9	Invalid menu choice	Invalid choice. Please enter 1 to 4.
8	4	Exit	Exiting program.

C Program

```
#include <stdio.h>
#include <ctype.h>

#define MAX 200

void countCharacters(void);
void printCharacters(void);
void countVowelsConsonants(void);
void readLine(char str[], int size);
```

```

int main(void)
{
    int choice;

    do
    {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Count characters in a string\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                countCharacters();
                break;
            case 2:
                printCharacters();
                break;
            case 3:
                countVowelsConsonants();
                break;
            case 4:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Please enter 1 to 4.\n");
        }
    } while (choice != 4);

    return 0;
}

void readLine(char str[], int size)
{
    int i = 0;
    fgets(str, size, stdin);

    while (str[i] != '\0')
    {
        if (str[i] == '\n')
        {
            str[i] = '\0';
            break;
        }
        i++;
    }
}

void countCharacters(void)
{
    char str[MAX];
    int count = 0;

    printf("Enter a string: ");
    readLine(str, MAX);

    while (str[count] != '\0')
        count++;

    printf("Number of characters = %d\n", count);
}

void printCharacters(void)

```

```

{
    char str[MAX];
    int i = 0;

    printf("Enter a string: ");
    readLine(str, MAX);

    printf("Characters are:\n");
    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants(void)
{
    char str[MAX];
    int i = 0, vowels = 0, consonants = 0;
    char ch;

    printf("Enter a string: ");
    readLine(str, MAX);

    while (str[i] != '\0')
    {
        ch = (char)tolower((unsigned char)str[i]);

        if (ch >= 'a' && ch <= 'z')
        {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
                vowels++;
            else
                consonants++;
        }
        i++;
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
}

```

Compilation and Execution

```

gcc -std=c99 hdoc36.c -o hdoc36
./hdoc36

```

Screenshot Spaces

Screenshot 1 - Successful compilation



Screenshot 2 - Choice 1: Count characters

```
— STRING OPERATIONS MENU —
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Enter a string: Hello
Number of characters = 5
```

```
— STRING OPERATIONS MENU —
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Enter a string: Hello World
Number of characters = 11
```

Screenshot 3 - Choice 2: Print each character

```
— STRING OPERATIONS MENU —
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Enter a string: CAT
Characters are:
C
A
T
```

```
— STRING OPERATIONS MENU —
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Enter a string: AB
Characters are:
A
B
```

Screenshot 4 - Choice 3: Vowels and consonants

```
— STRING OPERATIONS MENU —  
1. Count characters in a string  
2. Print each character on a new line  
3. Count vowels and consonants  
4. Exit  
Enter your choice: 3  
Enter a string: Education  
Vowels = 5  
Consonants = 4
```

```
— STRING OPERATIONS MENU —  
1. Count characters in a string  
2. Print each character on a new line  
3. Count vowels and consonants  
4. Exit  
Enter your choice: C Programming 123!  
Enter a string: Vowels = 3  
Consonants = 8
```

Screenshot 5 – Invalid / Exit

```
— STRING OPERATIONS MENU —  
1. Count characters in a string  
2. Print each character on a new line  
3. Count vowels and consonants  
4. Exit  
Enter your choice: 9  
Invalid choice. Please enter 1 to 4.
```

```
— STRING OPERATIONS MENU —  
1. Count characters in a string  
2. Print each character on a new line  
3. Count vowels and consonants  
4. Exit  
Enter your choice: 4  
Exiting program.
```